Class 8

For a game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, a strategy profile s^* is an NE if

$$u_i(s^*) \geq u_i(s_i', s_{-i}^*) \quad \forall i \in N.$$

Iterated elimination of strictly Dominated strategies

$$G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

$$\hat{G} = \langle N, (S_i')_{i \in N}, (u_i)_{i \in N} \rangle$$

where $S_i = S_i'$ for $i \neq k$

$S_k' = S_k \setminus \{s_k\}$ such that s_k is strictly dominated.

If an NE is removed then that NE must have s_k as it's k^{th} component

This cannot happen. Therefore, iterative elimination of strictly dominated strategies can not remove any NE from the game.

Say s^* is a new NE in the game $\hat{G}$, in the sense that s^* was not an NE in G.

$$u_i(s^*) \geq u_i(s_i, s_{-i}^*) \quad \forall i$$
$$\forall s_i \in S_i' \quad \text{---} (*)$$

$\exists j$ s.t. $u_j(s^*) < u_j(s_j, s_{-j}^*)$ ... (**)

for some $s_j \in S_j$.

$\forall$ $u_i(s^*) \geq u_i(s_i, s_{-i}^*)$ $\forall i \in N$

$\forall s_i \in S_i$ if $i \neq k$

$\forall s_k \in S_k'$ if $i = k$

$= S_k \setminus S_k'$

(*) and (**) implies $j = k$ $s_k = \bar{s}_k$

$\Rightarrow u_k(s^*) < u_k(\bar{s}_k, s_{-k}^*) < u_k(t_k, s_{-k}^*)$

where t_k strictly dominates $\bar{s}_k$.

$t_k \in S_k'$

and $u_k(s^*) < u_k(t_k, s_{-k}^*)$, so s^* can not be a NE of the game $\hat{G}$

Theorem Suppose G is a game and $\hat{G}$ has been obtained from G by applying iterated elimination of strictly dominated strategies then the set of Nash Equilibria in G and in $\hat{G}$ remain the same.

Iterative elimination of weakly dominated strategies

	L	R
T	0,0	2,1
B	3,2	1,2

(B, L) and (T, R)

	L	R
T	2,1	
B	1,2	

	R
T	2,1

In general for iterative elimination of weakly dominated strategies, some NE may get removed.

- $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, $G' = \langle N, (S_i')_{i \in N}, (u_i)_{i \in N} \rangle$

  where $S_i' = S_i \quad \forall i \neq k$

  $S_k' = S_k \setminus \{\bar{s}_k\}$ such that $\bar{s}_k$ is weakly dominated for player k.

Question: Can we get some new NE in G' that were not NE in G?

Mixed strategies

A	B
Rs 100	Rs 500 1/4
	Rs 0 3/4

We will assume rational players evaluate probability distribution by Expectation.

1 \ 2	L	R
T	(2,	(1
B	(4	(0

$\Leftrightarrow$

1 \ 2	L	R
T	(3,	(1.5,
B	(3.5,	(0.5

$(B,L) > (T,L) > (T,R) > (B,R)$

$G \triangleq \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle \Leftrightarrow \hat{G} = \langle N, (S_i)_{i \in N}, (v_i \circ u_i)_{i \in N} \rangle$

where $v_i : \mathbb{R} \to \mathbb{R}$ strictly increasing function

Now, arbitrary monotonic transformation is not allowed. We will only allow affine transformation

$$v_i = a_i u_i + b_i \quad \text{where } a_i > 0$$